

Q1. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

Given that $-4 \leq x \leq 3$, and $5 \leq y \leq 11$, the minimum value of which of the following expressions is the least?

- ☐ a) $x^3y - xy^3$
- ☐ b) $x^3y - xy^2$
- ☐ c) $x^2y + xy^2$
- ☐ d) $y^2x + x^3y$

Q2. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

In an isosceles triangle, each of the equal sides is 10 cm long. If the area of the triangle is less than 48 sq.cm, which of the following is not a possible measure (in cm) of the third side?

- ☐ a) 11
- ☐ b) 9
- ☒ c) 14 ✓ **Your answer is correct**
- ☐ d) 18

Q3. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

What is the domain of the function $f(x) = \frac{24}{12 - [x]}$, where $[x]$ is the greatest integer less than or equal to x ?

☐ a)

$(-\infty, 12) \cup (13, \infty)$

☐ b)

$(-\infty, 12] \cup [13, \infty)$

☐ c)

$(-\infty, 12) \cup [13, \infty)$

☐ d) $(-\infty, 12] \cup (13, \infty)$

Q4. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

A man rowing upstream in a river accidentally dropped his hat into the river. After travelling a distance of $3x$, he noticed his loss and immediately turned back to retrieve his hat. He caught up with his hat after it had covered a distance of $2x$. If the speed of the stream was k percent of the speed at which he rows in still water, find k .

Enter your answer value rounded off and accurate up to two decimal places.

Q5. DIRECTIONS *for questions 4 and 5:* Type in your answer in the input box provided below the question.

There are two arithmetic progressions, A_1 and A_2 , whose first terms are 7 and 5 respectively and whose common differences are 6 and 8 respectively. How many terms of the series are common in the first n terms of A_1 and A_2 , if the sum of the n^{th} terms of A_1 and A_2 is equal to 9000?

Q6. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.

If $5x$ is equal to $18y$, $25z$ is equal to $2y$, $60z$ is equal to $0.8u$, $51u$ is equal to $68v$, then all the following statements are true except

- ☐ a) $17x$ is equal to $10.2u$.
- ☐ b) $90y$ is equal to $1125z$.
- ☐ c) $36v$ is equal to $45x$.
- ☐ d) $28y$ is equal to $126v$.

Q7. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.

If three regular, unbiased dice are thrown, what is the probability that either a '2' or a '3' turns up on at least one of the dice?

☐ a)

$$\frac{7}{27}$$

☐ b)

$$\frac{19}{27}$$

☐ c)

$$\frac{64}{216}$$

☐ d)

$$\frac{37}{54}$$

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.

A natural number N is 100 times the number of its factors. Find the sum of the digits of N , given that N has only two prime factors.

Q9. DIRECTIONS *for question 9:* Select the correct alternative from the given choices.

The question below gives certain information, followed by two quantities, I and II. Compare the quantities I and II, and then select the option that gives the correct relationship between them.

A piece of work is carried out by a group of men, all of equal capacity, in such a way that on the first day one man works and on every subsequent day one additional man joins the work. A group of women, all of equal capacity is engaged to carry out a second piece of work with ten women starting the work on the first day and one woman leaving the work at the end of every day. The second piece of work requires three times the effort required for the first piece of work and each man is thrice as efficient as each woman. It is known that one man working alone can complete the first piece of work in 6 days.

I. Number of days in which the second piece of work is completed.

II. Number of days in which the first piece of work is completed.

- ☐ a) $I > II$
- ☐ b) $II > I$
- ☐ c) $I = II$
- ☐ d) **The relationship cannot be determined from the given information**

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

What is the number of distinct ways in which four squares can be chosen on a chessboard such that they all lie on the same diagonal?

Q11. DIRECTIONS for questions 11 to 14: Select the correct alternative from the given choices.

Find the sum of the first n terms of the following series:

$$\log x^2 + \log \frac{x^4}{y^3} + \log \frac{x^6}{y^6} + \log \frac{x^8}{y^9} + \dots$$

☐ a)

$$\log \left(\frac{x^{n+1}}{y^{3\left(\frac{n-1}{2}\right)}} \right)^n$$

☐ b)

$$\log \left(\frac{x^{n+1}}{y^{3\left(\frac{n+1}{2}\right)}} \right)^n$$

☐ c)

$$\log \left(\frac{x^{n+1}}{y^{3\left(\frac{n}{2}\right)}} \right)^n$$

☐ d)

$$\log \left(\frac{x^{n+1}}{y^{\left(\frac{n-1}{2}\right)}} \right)^n$$

Q12. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

Prateek has a cuboidal box, having a square base, and 14 identical wooden spheres with him. He observed that he could snugly and perfectly arrange exactly nine of the fourteen spheres at the bottom of the box in a single layer comprising three rows and three columns. He then placed another layer of four spheres, stably and symmetrically on top of the bottom layer (i.e., such that each sphere in the second layer touched exactly four spheres of the bottom layer). Finally, he placed the last sphere, stably and symmetrically, on top of the second layer and observed that he could then just close the lid of the box. Assuming that the thickness of the box is negligible, find the ratio of the height of the box to the radius of each sphere.

☐ a)

$$2(2\sqrt{3}-1)$$

☐ b)

$$2(\sqrt{2}+1)$$

☐ c)

$$2(\sqrt{3}+1)$$

☐ d)

$$3(2\sqrt{2}-1)$$

Q13. DIRECTIONS *for questions 11 to 14:* Select the correct alternative from the given choices.

D_1, D_2, D_3 and D_4 are recurring decimals given by $D_1 = 0.\overline{q_1q_2q_3q_4}$, $D_2 = 0.q_1q_2\overline{q_3q_4}$, $D_3 = 0.\overline{q_1q_2q_3q_4}$ and $D_4 = 0.q_1q_2q_3\overline{q_4}$, where q_1, q_2, q_3 and q_4 are four digits from 1 to 9. Which of the following numbers when multiplied by at least one of the above-mentioned decimals will result in an integer, for any values of q_1, q_2, q_3 and q_4 ?

- ☐ a) 1800
- ☐ b) 19980
- ☐ c) 198
- ☐ d) 1111

Q14. DIRECTIONS for questions 11 to 14: Select the correct alternative from the given choices.

The marked price and the cost price of a watch are in the ratio 4 : 3. The discount percentage offered before it was sold and the profit/loss percentage made on it are in the ratio 3 : 4. Find the profit/loss percentage.

☐ a)

$16\frac{2}{3}\%$ Profit

☐ b)

$8\frac{1}{3}\%$ Profit

☐ c)

$8\frac{1}{3}\%$ Loss

☐ d) **Cannot be determined**

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

If $7x + 11y = S$ has exactly 3 solutions, in which both x and y are positive integers, what is the minimum value of S ?

Q16. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.

All the natural numbers from 1 to 100 are written adjacent to each other in a single row. Now, this is considered as a single number, N , with 192 digits. N is then broken into two parts, each part being considered to be a new number, by placing a single partition between any two consecutive digits of N and the two numbers so obtained are added to form a new number, N_1 . Now, N_1 is broken into two parts in a manner similar to that followed above for N , and the two numbers thus arrived at are again added to form another new number, N_2 . This process is continued to arrive at N_3 , N_4 ,.... and so on, till we finally get a single digit number N_m . What is the value of N_m ?

- ☐ a) 5
- ☐ b) 3
- ☐ c) 1
- ☐ d) **Depends on the actual order of partitioning the original number N .**

Q17. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.

Given $f(x) = a|x| - bx^2$, where a and b are constants. Then at $x = 0$, $f(x)$ is

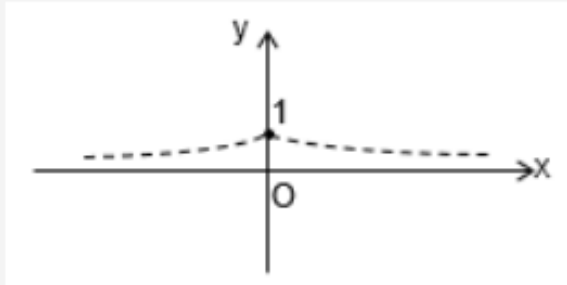
- ☐ a) maximized whenever $a > 0, b > 0$.
- ☐ b) minimized whenever $a > 0, b > 0$.
- ☐ c) minimized whenever $a < 0, b > 0$.
- ☐ d) maximized whenever $a < 0, b > 0$.

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

You have 80 pearls. One is lighter than the remaining. What is the minimum number of weighings required on a common balance to ensure that the odd pearl is identified?

Q19. DIRECTIONS for questions 19 and 20: Select the correct alternative from the given choices.

Which of the following functions best corresponds to the graph shown below?



☐ a)

$$y = \frac{1}{\ln x}$$

☐ b)

$$y = \frac{1}{e^x}$$

☐ c)

$$y = \frac{1}{e^{|x|}}$$

☐ d) $y = e^{|x|}$

Q20. DIRECTIONS for questions 19 and 20: Select the correct alternative from the given choices.

Find the sum of the series $S = \frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{2}+2} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{2+\sqrt{6}} + \dots + \frac{1}{\sqrt{119}+\sqrt{121}}$.

☐ a) 5

☐ b)

$$\frac{2\sqrt{15}+1}{2}$$

☐ c)

$$5 + \frac{2\sqrt{15}-1}{\sqrt{2}}$$

☐ d)

$$\frac{10+2\sqrt{30}+\sqrt{2}}{2}$$

Q21. DIRECTIONS *for question 21:* Type in your answer in the input box provided below the question.

Five friends P, Q, R, S, T attended a party. If no two of them arrived at the same time and P arrived before Q, who, in turn, arrived before R, in how many different ways could they have arrived at the party?

Q22. DIRECTIONS *for question 22:* Select the correct alternative from the given choices.

A cat saw a squirrel when the latter was exactly at the middle of a vertical pole. At that moment, the angle of elevation of the squirrel, as observed by the cat, was 30° . The cat then ran a certain distance towards the pole and chased the squirrel, which then ran to the top of the pole. The cat, which is now 20 m away from the foot of the pole, finds that the squirrel now has an angle of elevation of 60° . What distance did the cat run?

☐ a) 6 m

☐ b) 10 m

☐ c)

$10\sqrt{3}$ m

☐ d)

$6\sqrt{3}$ m